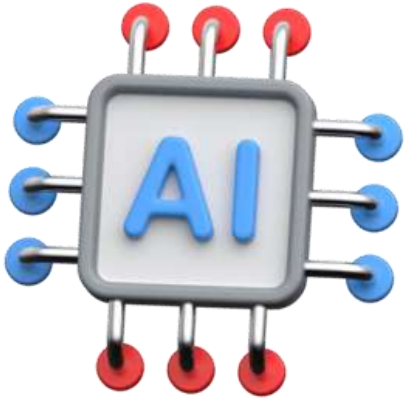
SECTION - B



Advanced Concepts of Modeling in AI

Unit 2

Learning Objectives:

1. To familiarize students with supervised, unsupervised and reinforcement learning based approach.
2. To Introduce students to the neural network.

Learning Outcomes:

1. Understand supervised, unsupervised and reinforcement learning based approach.
2. Understand Neural Networks.

Key-concepts:

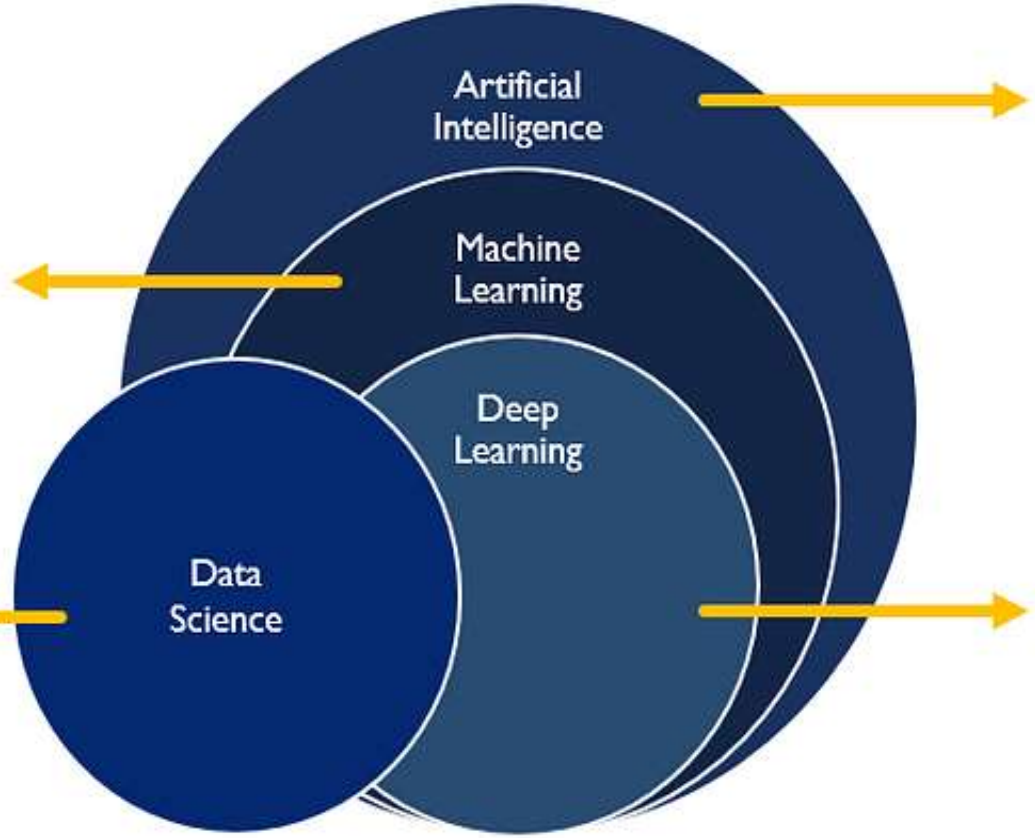
1. Supervised, unsupervised and reinforcement learning based approach
2. Neural Networks

2.1 Revisiting AI, ML, DL

To build an AI based project, we need to work around Artificially Intelligent models or algorithms. This could be done either by designing your own model or by using the pre-existing AI models. Before jumping into modelling let us clarify the definition of Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL).

Differentiate between AI, ML and DL

Machine Learning is a subset of AI, and builds a model based on training data to make predictions



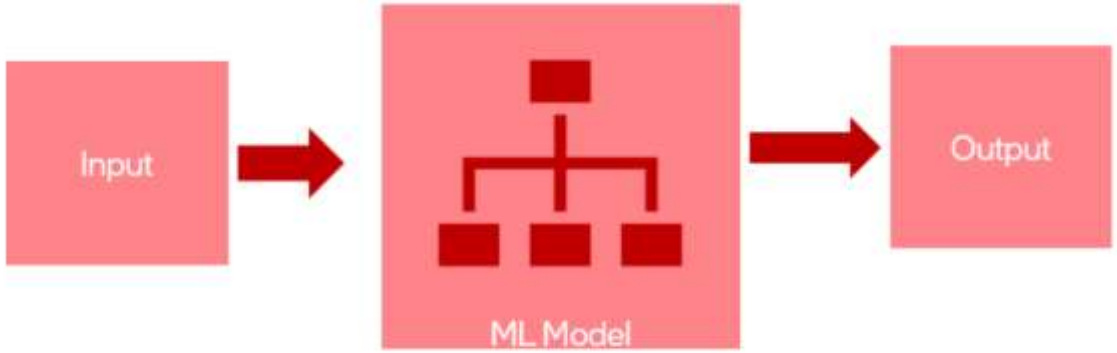
Artificial Intelligence means creating smart machines to mimic human behavior

Data Science is a subset of AI. It is an area of statistics, scientific methods, etc. to extract meaning and insights from data..

Deep learning is a subset of ML, a class of ML algorithms to solve complex problems.

Machine Learning (ML)

Machine Learning (ML) is a way for computers to learn from data and improve their performance without being told exactly what to do at every step. Instead of following only fixed instructions, the computer looks for patterns in data, uses them to make predictions or decisions, and gets better over time by learning from new data and its mistakes.

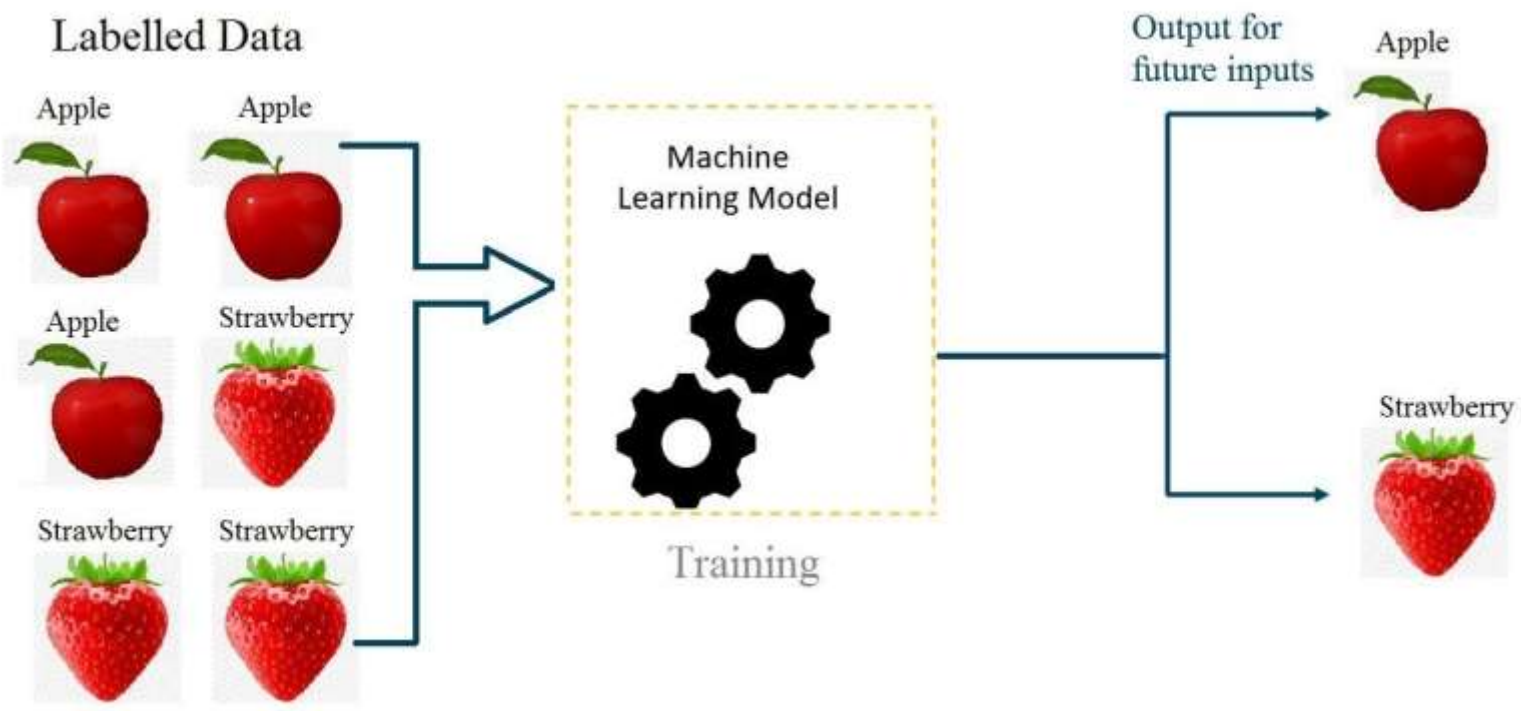


Block Representation – Machine Learning (ML)

This is just a broad representation of how a machine learning model works. Input (past or historical data) is given to the ML model and the model generates output by learning from the input data.

How does it work?

Example shows labelled images (every image is tagged either as apple or strawberry) are given as input to the ML model. ML model learns from the input data to classify between apples and strawberries and predicts the correct output as shown.



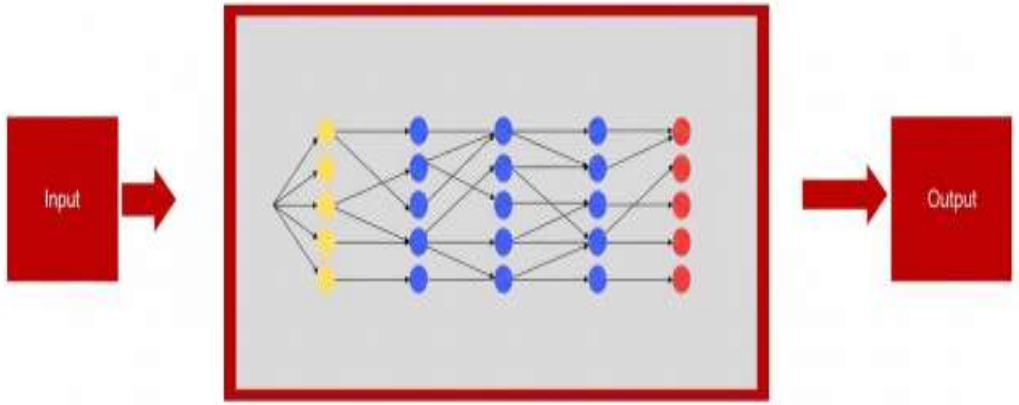
Example - ML

Common Machine Learning Examples and use Cases:

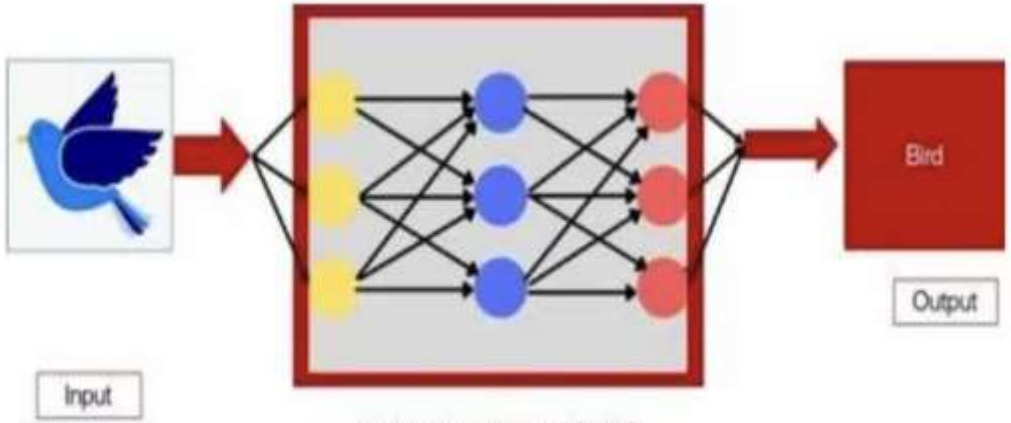
1. **Product/Streaming Recommendations**: Algorithms (like on Spotify or Netflix) analyze browsing/listening history to suggest new content.
2. **Voice Assistants**: Systems like Siri, Alexa, and Google Assistant use natural language processing (NLP) to understand commands and speech.
3. **Fraud Detection**: Banks automatically flag suspicious transactions using pattern recognition.
4. **Email Filters**: Classifying incoming emails as spam or legitimate, learning from user behavior.
5. Transportation/Navigation: Google Maps analyzes traffic to determine the fastest routes.
6. **Healthcare Diagnostics**: Analyzing X-rays or genetic data to detect diseases earlier than traditional methods.
7. **Predictive Text/Autocorrect**: Smart keyboards learn typing patterns to suggest words.
8. **Chatbots**: AI-powered customer service bots handling support queries 24/7.

Deep Learning (ML)

Deep Learning (DL) trains systems using large amounts of data, enabling them to learn patterns and improve on their own. These systems can even create their own algorithms, making DL the most advanced form of AI. Following is the block diagram of deep learning:



Artificial Neural Network (ANN)

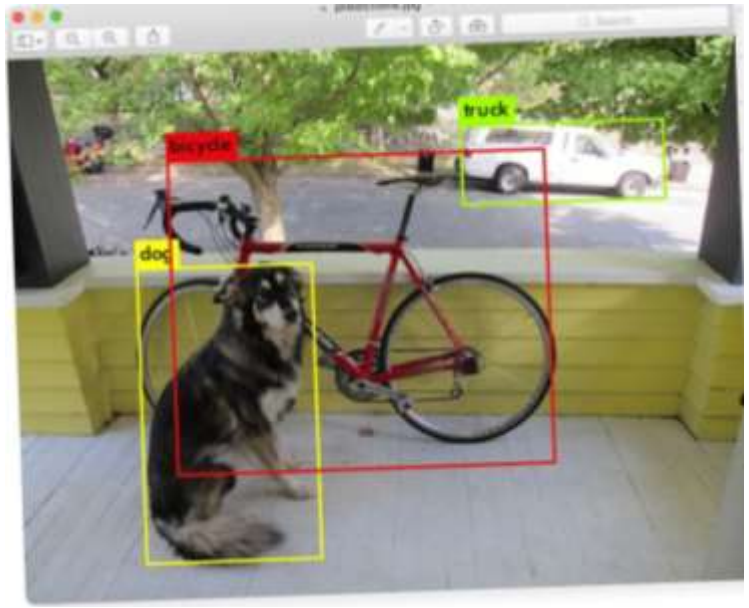


Artificial Neural Network(ANN)

Here is an example which shows pixels of a bird image given as input to the DL Model and the model is able to analyze and correctly predict that it is a bird using a deep learning algorithm (ANN).

Example - DL

- ✓ **Object Identification** – Object classification in deep learning tackles the task of identifying and labeling objects within an image. It essentially uses powerful algorithms to figure out what's in a picture and categorize those things.



- ✓ **Digit Recognition** – Digit recognition in deep learning tackles the challenge of training computers to identify handwritten digits (0-9) within images.